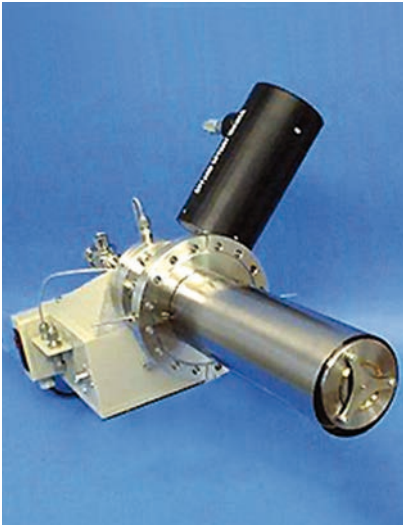
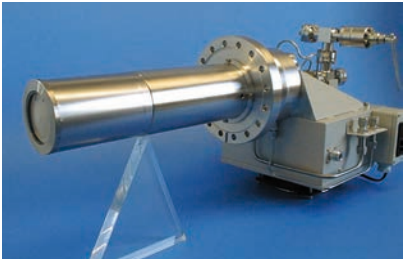


MBE SPECIFIC SOURCES

The MBE 49 production friendly architecture accommodates field proven components for superior reliability. The following table is an overview of the MBE material/source capabilities.

MATERIALS	CELL SPECIFICATION	CELL MODEL
Ga, In , Al	Standard effusion cell	ABN 700 DF
Ga, In	High capacity effusion cell	ABI 1000
As	Valved cracker cell	VAC 2000
P	Valved cracker cell	KPC 1200
Sb	Valved cell	VCOR
Nitrogen	Ammonia delivery module	HTI 440
Nitrogen, Oxygen, Hydrogen	RF plasma source	RF – X- 50/100 X : N, O, H
Diluted nitride	Valved plasma source	VRF-N-600
La, Hf, Ti, Fe	Ebeam evaporator unit	2x100 cc capacity 4x40 cc capacity

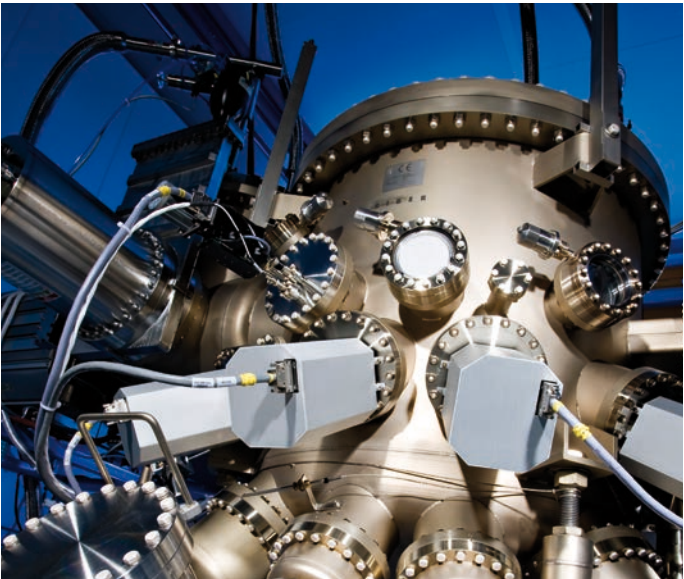


The MBE 49 flexible architecture allows to easily accommodate different configurations such as loading/ unloading chambers, one process and/ or analysis chamber, with up to two growth chambers, in a clusterized design version.

BENEFITS

The MBE 49 mass production 4” systems benefits of more than 10 years and of 42 systems installed worldwide experience. The MBE 49 conception is ruled by industry latest standards and is delivers the highest performances on the market today. With our MBE 49 you will benefit of:

- Superior throughput
up to 15,100 4”/year
- Longest growth campaign
- Very low defect density
≤ 40 cm-2 on GaAs
- Excellent uniformity
≤± 1% thickness uniformity GaAs
≤± 1% composition uniformity GaAlAs
- Day to day highest reproducibility
- Full automation – batch to batch process
- Ease of use and maintenance
- Riber worldwide support



TECHNOLOGICAL LEADERSHIP

Riber is the world leading supplier of MBE processing equipments and related services.

The company has reached its 800th system, with at least one system in each of the 35 countries involved with MBE, which represents 75% of market worldwide. Capitalizing on 30 years of experience, the core philosophy of the company is to design systems in close association with customers. Riber invented and designed major features which are now found in all MBE systems. Riber plays a key role in the development of MBE technology, providing customers with solutions from equipment to epitaxial growth.

REFERENCES

- Skyworks – USA

IntelliEpi – USA

IQE – Singapour

Century Epitech – China

Raytheon – USA

Oclaro – Switzerland

Thales Research & Technology – France
- Innolume – Germany

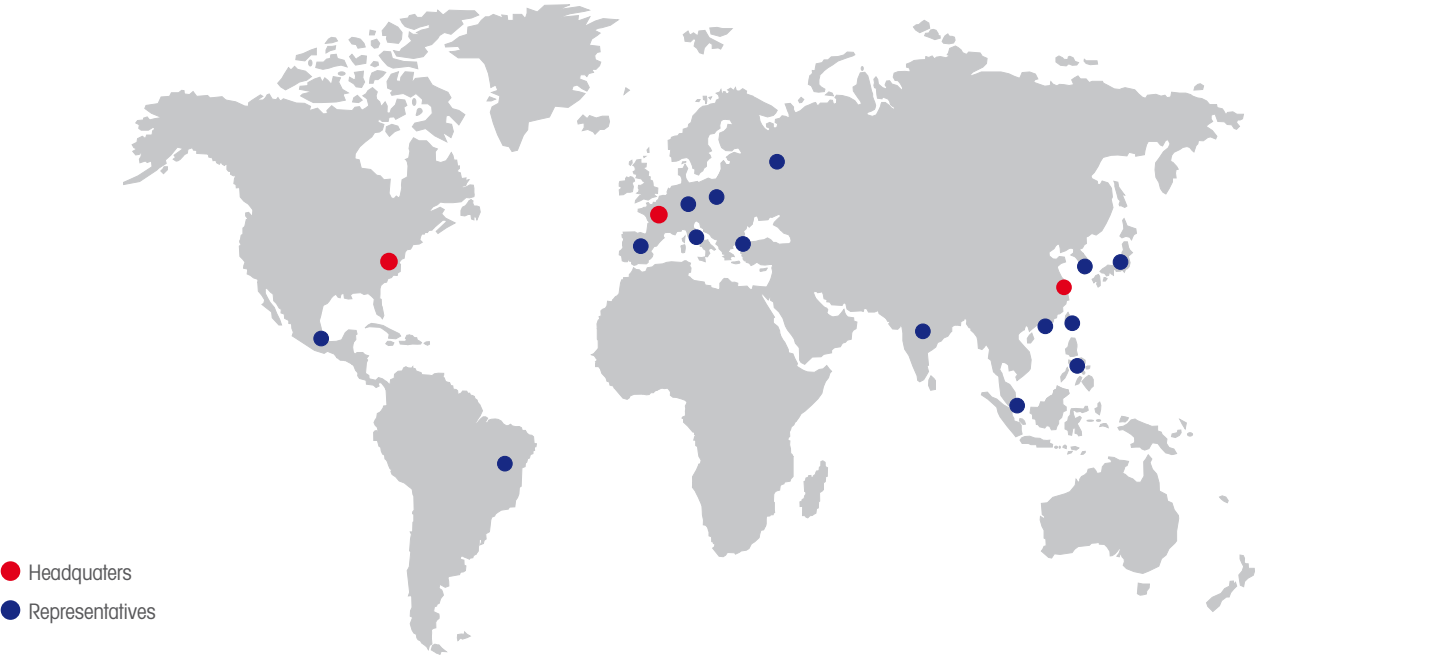
Asahi Kasei – Japan

Picogiga / Soitec – France

Japanese Companies

IMEC – Belgium

WORLDWIDE PRESENCE



For more information:
Tel: +33 (0)1 39 96 65 00 / Email: info@riber.com /Internet: www.riber.com

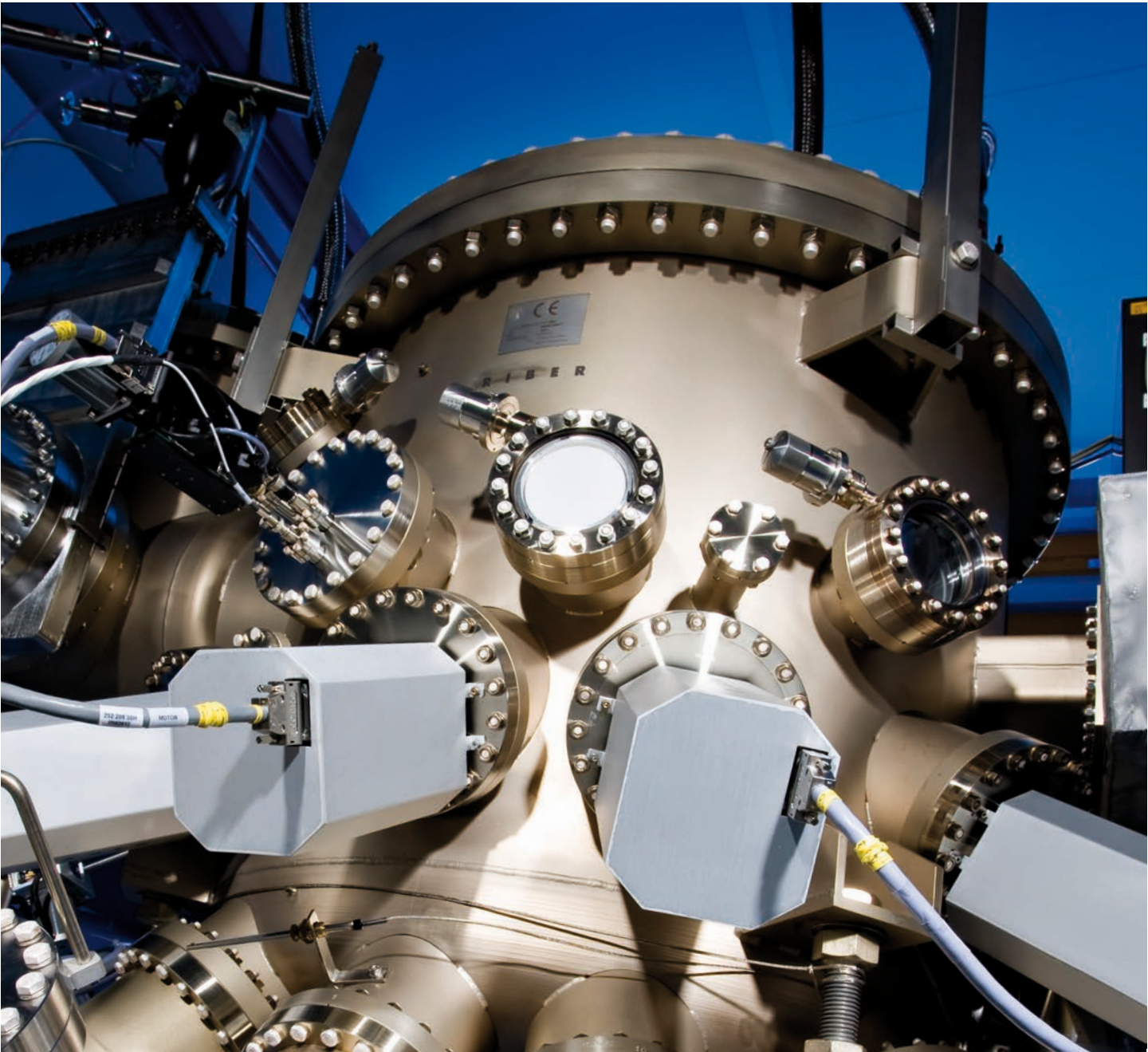
MBE SYSTEMS AND COMPONENTS FOR R&D AND PRODUCTION

MBE 49
BRINGING YOU THE EXCLUSIVE PRODUCTION SOLUTION

- TURNKEY MBE SYSTEM
- 4X4” WAFER CAPACITY PER RUN
- LARGEST RANGE OF APPLICATIONS
- UP TO 30 PLATENS LOADING CAPACITY

INNOVATIVE SOLUTIONS FOR SEMICONDUCTOR INDUSTRY

RIBER



THE HIGHEST STANDARDS OF QUALITY AND RELIABILITY FOR MASS PRODUCTION OF EPIWAFERS

A peace of mind running for the compound semiconductor devices

In 1998, Riber launches the first fully automated MBE production system, the MBE 49. This multi-3-inch to 6-inch wafers equiptool sets the industry standards in terms of material quality, yield, throughput and low operation and maintenance costs. With 40 systems processes worldwide, the MBE 49 is definitively considered by the MBE community to be the reference for both in-house manufacturers and merchant suppliers of optoelectronic, microwave electronic and detector device epiwafers.

The MBE 49 production tool system provides all the necessary components, pre-growth treatment, in-situ characterization capabilities. Designed for volume production of electronic and optoelectronic device structures, the MBE 49 system also meets requirements for new process, such as Gallium Nitride, Oxide, silicon based ... applications.

To increase flexibility, the MBE 49 is easily integrated in a fully automated clusterized architecture.

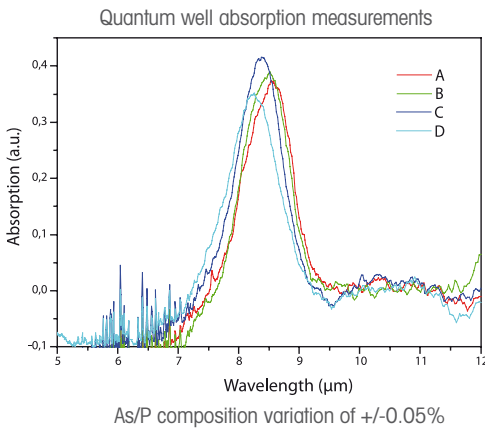


PERFORMANCE

The MBE 49 multiwafer production system equipped with components at the technical edge delivers the highest reachable performances on the market.

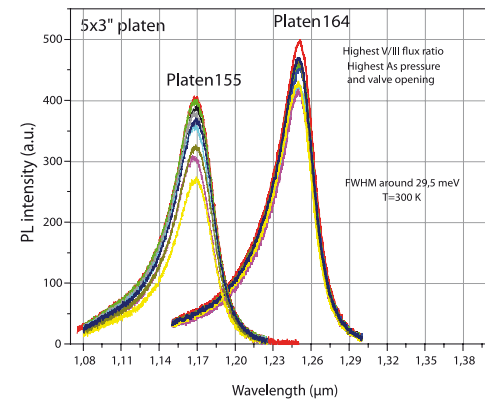
Run to run reproducibility better than industrial specifications

Growth consists in QWIP structures of GaAs wells embedded between AlGaAs barriers on 3" GaAs substrates. Reproducibility measurements were carried out over a batch a 12 platens (5x3-inch), corresponding to 96 hours growth. Run to run reproducibility results:
± 0.15Å of GaAs quantum well thickness variation
± 0.2% of Al mole fraction variation over the AlGaAs barriers
± 0.17µm of wavelength variation on four wafers



As/P Composition variation of ±0.05%

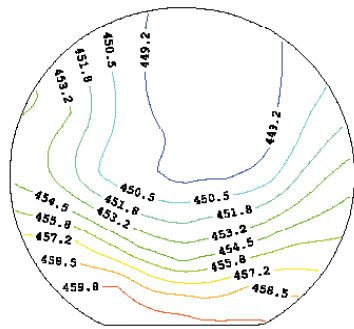
- Growth of InAsP/InP multi-quantum wells using the VAC 2000 arsenic cracker cell and KPC 1200 phosphorus cracker cell.
- Composition of the InAsP quantum well is highly dependant of the As/P flux ratio and of the substrate temperature over the platen.
- Photoluminescence measurement over the 5x3" platen shows ± 4Å wavelength variation at the optimal As/P flux rate corresponding to a As/P composition variation of +/- 0.05%.



Photoluminescence spectra over 2 platens (155 and 164) of InGaAs/InP multi quantum wells

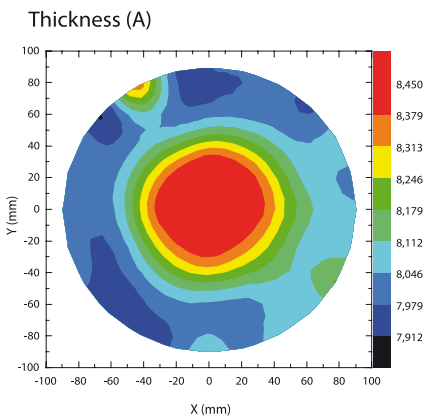
Excellent GaN uniformities.

- 4" Si and 3" SiC:
Thickness and composition < ± 1%
Sheet resistance < ± 1%
- Wafer to wafer on one platen: 3x4" Si and 5x3" SiC
Thickness and Composition < ± 1%
Sheet resistance < ± 1%
- Run to run reproducibility:
Thickness and Composition < ± 2%



State-of-the-art quality of high K material on SiO2/Si

- 8nm thin LaAlO3 layer deposited at 300°C under molecular oxygen (4.10-6 Torr) on a 1.1nm SiO2/Si 8-inch wafer.
- ± 2% thickness variation over the sample by ellipsometry XRR confirms structural quality of the oxide layer. The density of LaAlO3 extracted by XRR is 82% of the bulk density.



Ellipsometry mapping of 8nm of Al2O3 deposited on 1.1nmSiO2/Si 8" wafer.

PRODUCT GUIDE

the MBE 49 multiwafer production system is available for the four main applications present on the market today. Systems are all equipped with the necessary equipment and characterization tools adapted to their specific application
For each MBE 49 version, teams of MBE expert provide a process and technical support to users.
The following gives an overview of the main features of our MBE 49 models.

MBE 49 GaAs

The reference

- Dedicated Arsenic and Antimony valved cracker sources
- Up to twelve source ports
- Unique fast and soft actuation source shutter actuation for complex QCL structures
- User friendly automation interface and production process dedicated software



MBE 49 InP

The phosphorus handling production system

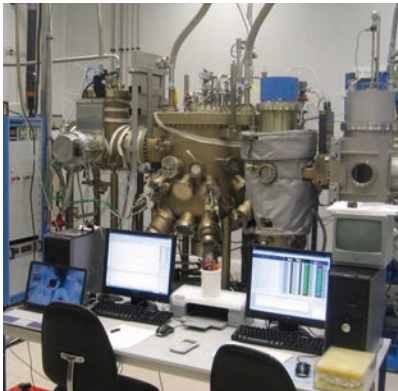
- Proven design for phosphorus based applications
- Field proven phosphorus recovery procedure
- Patented 3-zones Phosphorus valved cracker cell
- User friendly automation interface and production process dedicated software



MBE 49 GaN

Unique GaN production system

- Dedicated Ammonia injector and/or Nitrogen plasma source
- Efficient Gallium and Aluminium sources
- Process temperature between 650-950°C
- Reliable and optimized Ammonia pumping and recycling operations
- User friendly automation interface and production process dedicated software



MBE 49 OX

Field proven system for the next generation of CMOS

- Multi pocket E-beam evaporator unit
- Dedicated high temperature effusion cells
- High stability / reproducibility through selective beam detection through RGA or/and QCM
- Wafer treatment up to 1000°C
- Process temperature uniformity from 400 -1000°C
- User friendly automation interface and production process dedicated software

